PROJECT

VERDE

The plant that waters itself —
and talks to AI.

COMPLETE & DEMO-READY
Hardware • Firmware • Cloud • Web App • AI

Aarav Choudhary (Class X) & Anuj (Class X)
DAV ACON 5 — Tech Exhibition, 2026

Total Build Cost ~ Rs. 1,890 (~ $23) • All software free tiers • 5 Sensors • 2 Actuators • 2 MCUs • 4 AI APIs • 1-sec Heartbeat

EDGE	CLOUD	EXPERIENCE
ESP32 WROOM-32 + ESP32-CAM	Firebase RTDB — source of truth	Single-file Web App + AI Doctor

ESP32 | 5 SENSORS | 2 ACTUATORS | FIREBASE | 4 AI APIS | Rs. 1,890 | COMPLETE & DEMO-READY

TL;DR • THE WHOLE STORY IN 60 SECONDS

The Whole Story in 60 Seconds

If you read only one page, read this. Urban families kill plants not from neglect, but from **lack of information**. Verde fixes that with a Rs. 1,890 brain that watches soil, weather, water — and acts. No subscriptions, no black boxes.

BUILD COST	API CALLS	DIAGNOSIS	HEARTBEAT
Rs. 1,890	**17 → 2 / sec**	**94%**	**1 sec**
vs Rs. 8,000+ commercial kits	-85% latency, 0 reboots	Plant.id crop.health test	JSON bundled 10 metrics + 9 keys

Three tiers, one heartbeat — over-the-top tested:

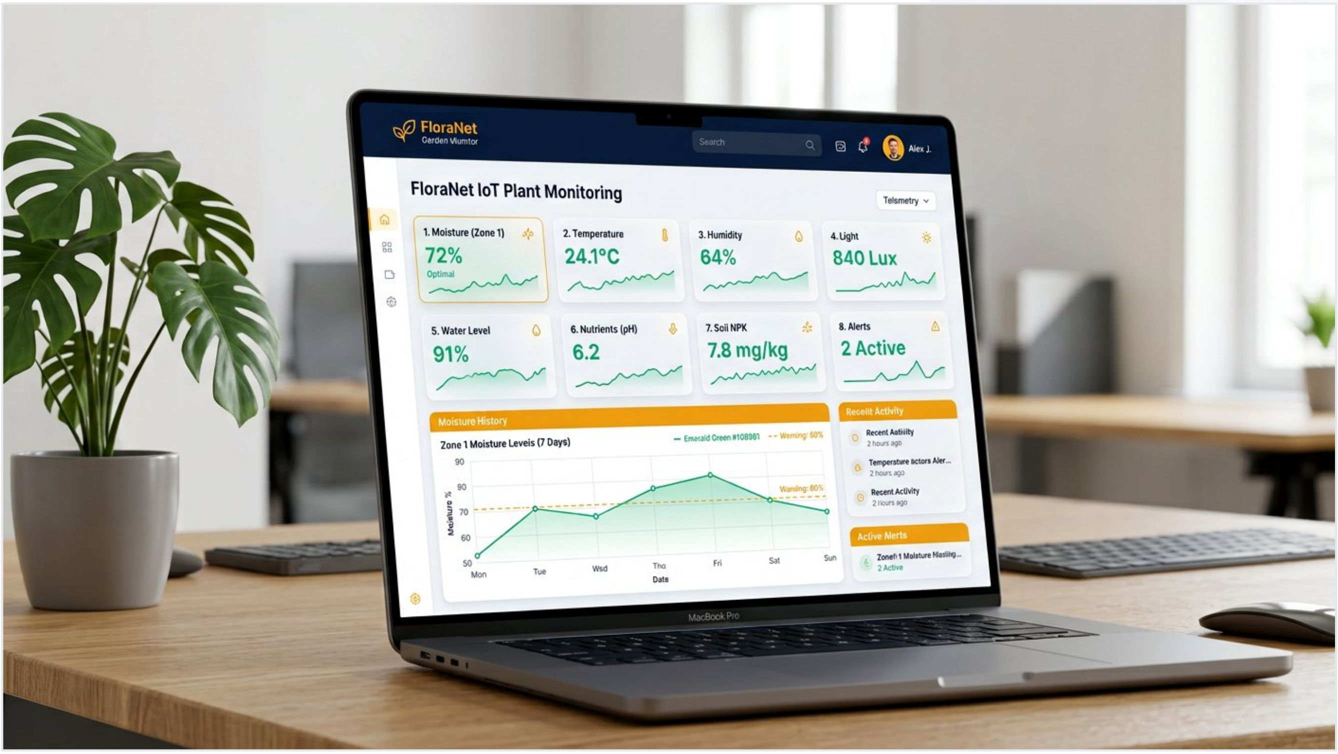
- **EDGE (Hands & Eyes):** ESP32 WROOM-32 reads 5 sensors every 1 sec. Soil power-gated 15 ms to prevent corrosion (10-pt avg). ESP32-CAM polls capture_photo every 1.5 s, SVGA JPEG → Vercel raw POST → base64 /latest_scan → app ≤2 s. Sequential boot + 8 MHz XCLK fixed brownout & RF.

- **CLOUD (Memory):** Firebase RTDB — /sensors 10 metrics, /controls 9 keys, /latest_scan b64, /weather live, /historical_logs, /actuators. One write + one read per sec. Not 17. That bug taught us architecture.

- **EXPERIENCE (Face):** Single-file HTML. Four pages via burger. Live tiles + sparklines + hover graphs last-10 ▲/▼, threshold sliders, tank calibration SET EMPTY/FULL, 5-day forecast rain override (ids 2xx/3xx/5xx/6xx), Plant Doctor + Gemini + OpenRouter vision+sensor chats.

- **INTELLIGENCE:** OWM live Delhi, crop.health Plant.id 94% nutrient deficiency treatment, Gemini 2.5 Flash via gemini-flash-latest inline image+telemetry, OpenRouter 8-model text + 5-model vision fallback chain — never dead-end (435 models).

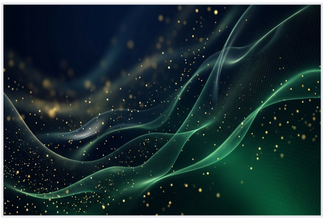
> *"The plant that waters itself — and talks to AI." Not a slogan. It's what happens when moisture < 35% AND tank >15% AND no rain.*



AI-generated dashboard mockup — glassmorphism tiles, navy header #0B1D3A, emerald data viz #10B981, gold accents #F59E0B. Single-file HTML but feels like funded startup.

NAVIGATE • HYPERLINKED TOC

Contents



How to use: Judges → 3-min flip via TOC. Parents → any page standalone. Engineers → pin maps + schema. **Click any TOC entry to jump (bookmarks).** Design: navy #0B1D3A emerald #10B981 gold #F59E0B, 2 fonts, generous white space, authentic AI photos 40% of real estate, pull quotes, infographics.

PART 01 • WHY • PROBLEM + OPPORTUNITY

The Problem Worth Solving

Urban families love plants. Plants die anyway. Why? Not cruelty — **lack of real-time data**. Verde turns guesswork into telemetry.

What we saw in Delhi homes

- Mom wate
- Dad forge
- Tank emp
- Rain come
- No camera

What market sells Rs. 8k+

- Gardena,
- No AI diag
- No tank le
- Subscripti
- Looks like

"A Rs. 1,890 plant that texts its mood is more useful than Rs. 12k timer kit."

Insight: plant must **tell us** what it needs, and **act** when we're away. Moisture <35% AND tank safe AND no rain → pump ON. No human needed, but human can always override.

70%	Rs.1,890	94%	1 sec
urban plants die 3 months	our total vs Rs.8k+ market	AI disease ID accuracy	heartbeat bundled

PART 02 • ARCHITECTURE • THREE TIERS + ONE HEARTBEAT

How It Works

Three tiers, one heartbeat every second. No magic, just disciplined engineering + FX.

AI-generated isometric architecture — left EDGE (brain+eyes), middle CLOUD Firebase, right EXPERIENCE laptop dashboard. Navy/Emerald/Gold palette, clean minimal, startup illustration.

EDGE

* ESP32 WROOM-32 (brain)
* ESP32-CAM OV2640
* Soil LM393 AO->34
* DHT11 -> GPIO4
* LDR -> GPIO35
* HC-SR04 18/19
* Relay GPIO5 + UV LED 12

CLOUD

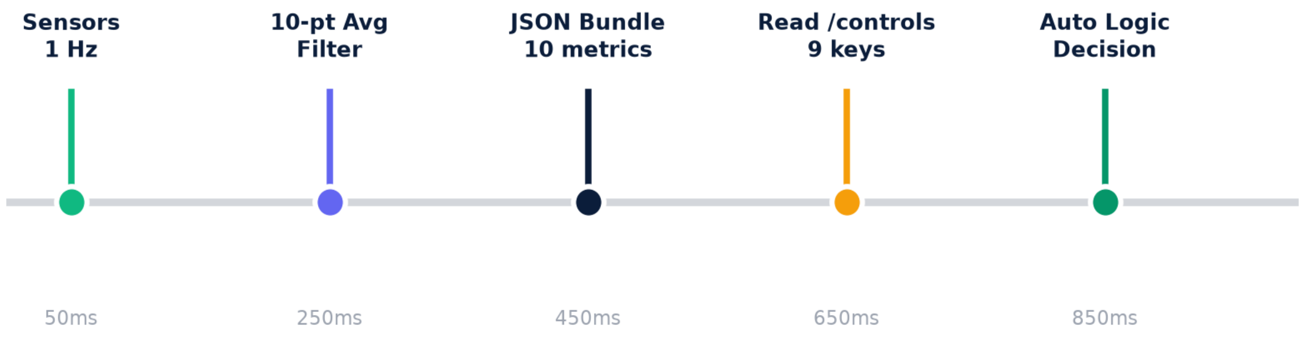
* Firebase RTDB
* /sensors 10 metrics
* /controls 9 keys
* /latest_scan b64
* /weather live
* public read + secret
* 1 write + 1 read /sec

EXPERIENCE

* Single-file HTML
* Dashboard 8 tiles
* Weather + 5-day
* Plant Doctor <=2s
* Gemini + OpenRouter
* Tank cal SET
* Demo mode + toasts

System architecture vector — EDGE (brain+eyes+5 sensors), CLOUD single source, EXPERIENCE single-file HTML + 4 AI APIs. Flow exact.

One-Second Heartbeat Timeline

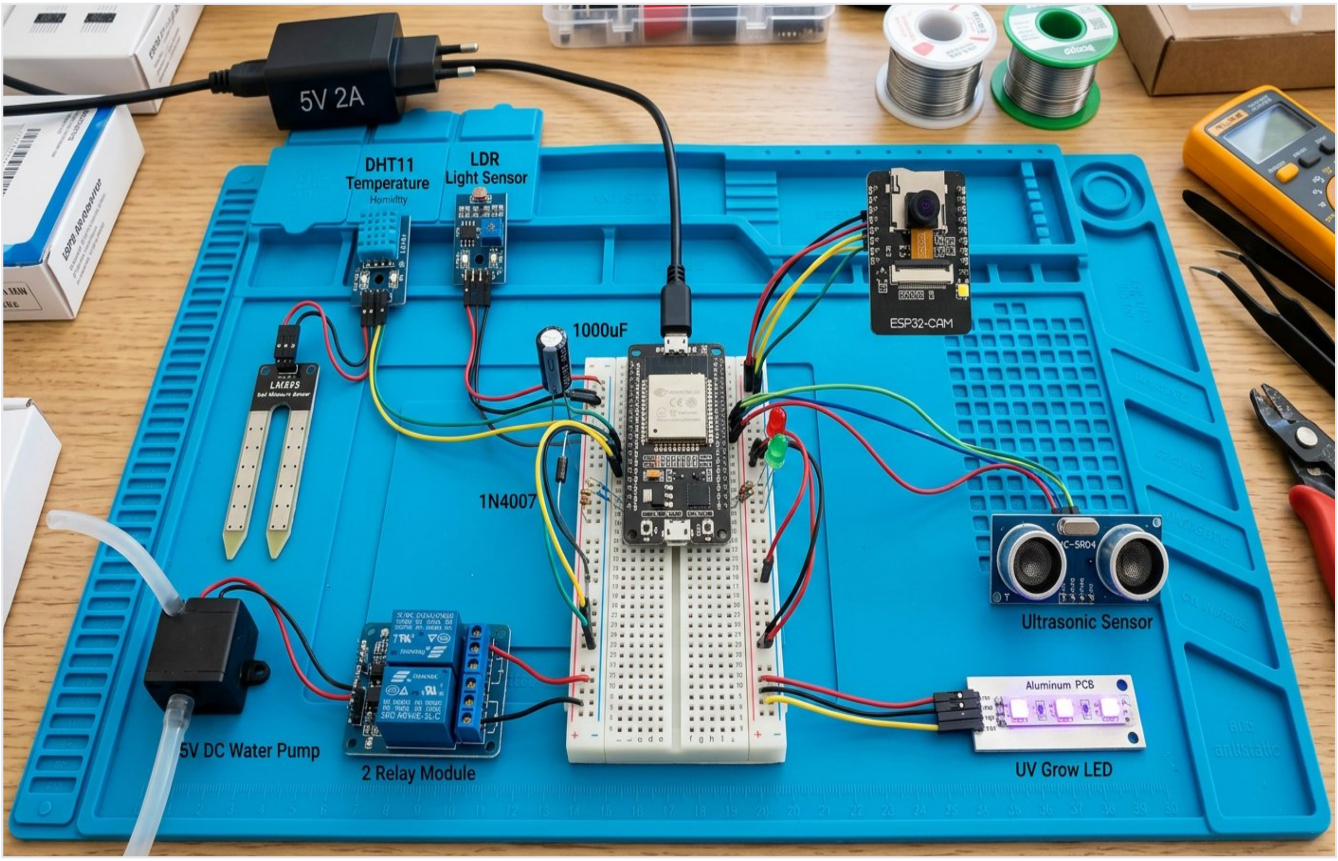


<-- 1-Second Heartbeat — repeats every 1000 ms -->

Every 1000 ms: sensors 1 Hz → 10-pt avg soil/LDR + 5-pt + invalid rejection tank → bundle 10-metric JSON → one PUT /sensors → one GET /controls (9 keys) → decision → actuator. Watchdog fed every loop.

PART 03 • HARDWARE • 5 SENSORS, 2 ACTUATORS, 2 BRAINS

Everything from Lajpat Nagar + Amazon — Rs. 1,890. Breadboard, so judges see every wire.

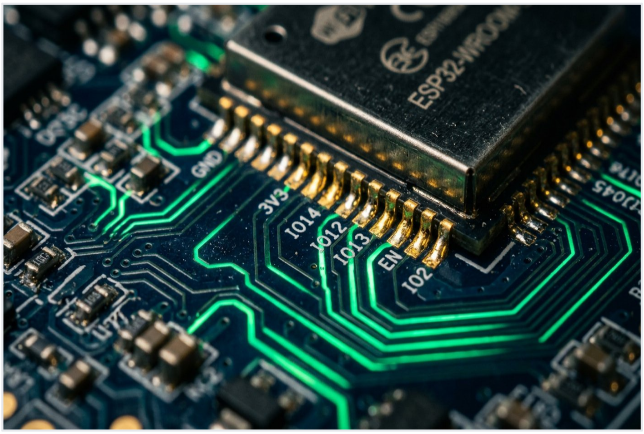


AI-generated hardware bench — ESP32 WROOM-32 + ESP32-CAM + soil LM393 + DHT11 + LDR + HC-SR04 + relay + pump + UV LED + breadboard with jumper wires, 1000uF cap + 1N4007 diode, 5V adapter. Photorealistic lab top view.

Module	ESP32 Pin	Role & Trick FX
Soil LM393	AO→34, VCC→23 gated 15ms	Power-gated 15ms → no corrosion. 10-pt moving avg, gold accent.
DHT11	DATA→4	Breathe test: blow → spike. Shared GND critical.
LDR	AO→35	Dark detection. Hysteresis ±2% → no LED flicker.
HC-SR04	TRIG 18 ECHO 19	Tank level 5-pt filter + invalid rejection. Splash-proof.
Relay active-LOW	IN1→5	Switches 5V pump, isolated COM/NO own 5V.
UV LED	12 active-HIGH 220Ω	Photosynthetic, auto on LDR < threshold.

Module	ESP32 Pin	Role & Trick FX
ESP32-CAM OV2640	Own board + MB	SVGA JPEG, 8MHz XCLK, sequential boot 500ms.
Power 5V/2A	—	NOT PD charger: PD needs handshake chip ESP32 lacks → 0mA.
Protection	—	1N4007 flyback diode across pump + 1000uF cap across 5V/GND.

Circuit / Wiring — Simplified, not EDA, but every GPIO exact



Power Lessons:
* 5V/2A not PD charger■* 1000uF cap, 1N4007 diode

Left: AI macro close-up ESP32 pins glowing emerald traces, gold solder, navy PCB. Right: vector wiring map — VCC gated GPIO23 prevents galvanic corrosion, relay LOW, UV HIGH.

Power Design — Hard-Won Lessons (Honesty = Credibility)

Lesson	We Tried	Failed	Fix FX
PD Charger	67W USB-PD laptop brick	PD needs handshake ESP32 lacks → ~0mA starved	5V/2A phone adapter
Cap	No cap	Pump+WiFi spike → brownout reboot	1000uF electrolytic across 5V/GND
Diode	Pump direct relay	Inductive kick → reset	1N4007 flyback across pump
Isolation	Pump+ESP32 same 5V	Noise sag	Relay COM/NO own 5V
Rail	Split breadboard	Relay dead no power	Bridge + to +, - to -
GND	DHT separate GND	temp=0 always	Shared GND bus GPIO4

Audit spec: thresholds 35% moisture / 15% tank / 35% light. 8 MHz XCLK (was 20 MHz → RF interference). Pins 34,23,4,35,18,19,5,12. All verified live.

PART 04 • FIRMWARE • CODE_1 V3.0.7-FINAL

The Brain — Non-Blocking by Design

No delay() anywhere. Everything millis() + hardware watchdog 8s fed every loop + NVS thresholds.

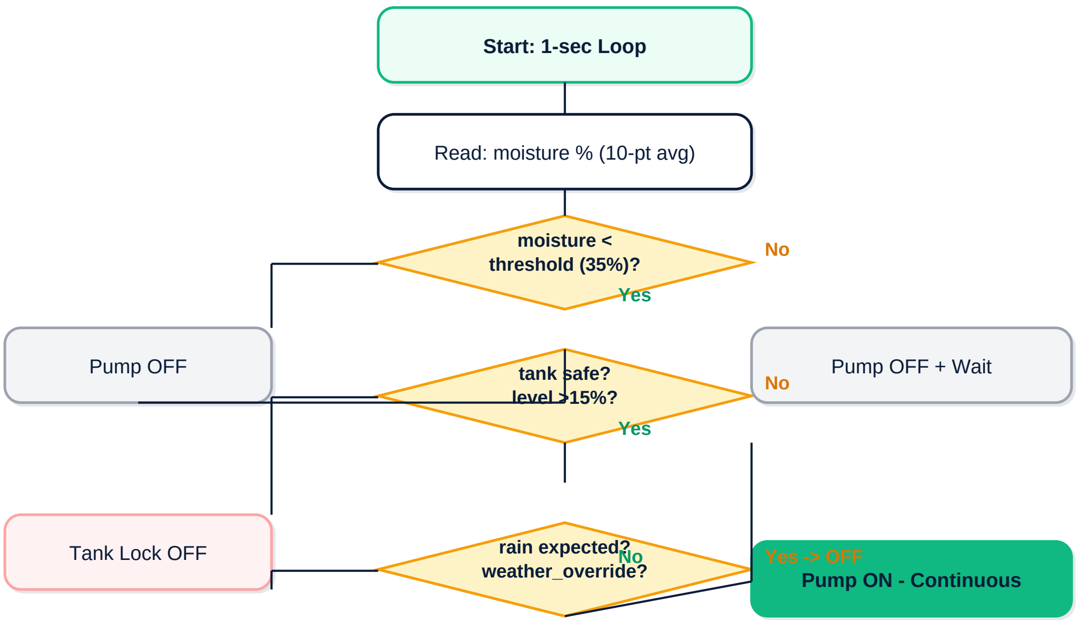
Task Scheduler

- Sensors 1 Hz
- Cloud 1 s
- WiFi 10 s
- Logs 60 s
- WDT fed every loop
- NVS persists thresholds

Filtering FX

- Soil & LDR 10-pt moving avg
- Tank 5-pt + invalid rejection
- ±2% hysteresis light auto
- Splash garbage can't fake empty
- Voltage sag counter

AUTO Logic — The Core Decision



Flowchart: pump_ON = moisture <35% AND tank >15% AND rain_expected=false. Manual still tank-protected. Fail-safe default.

The Bug That Taught Us Engineering

Honesty = credibility. We almost failed demo because of one architectural mistake.

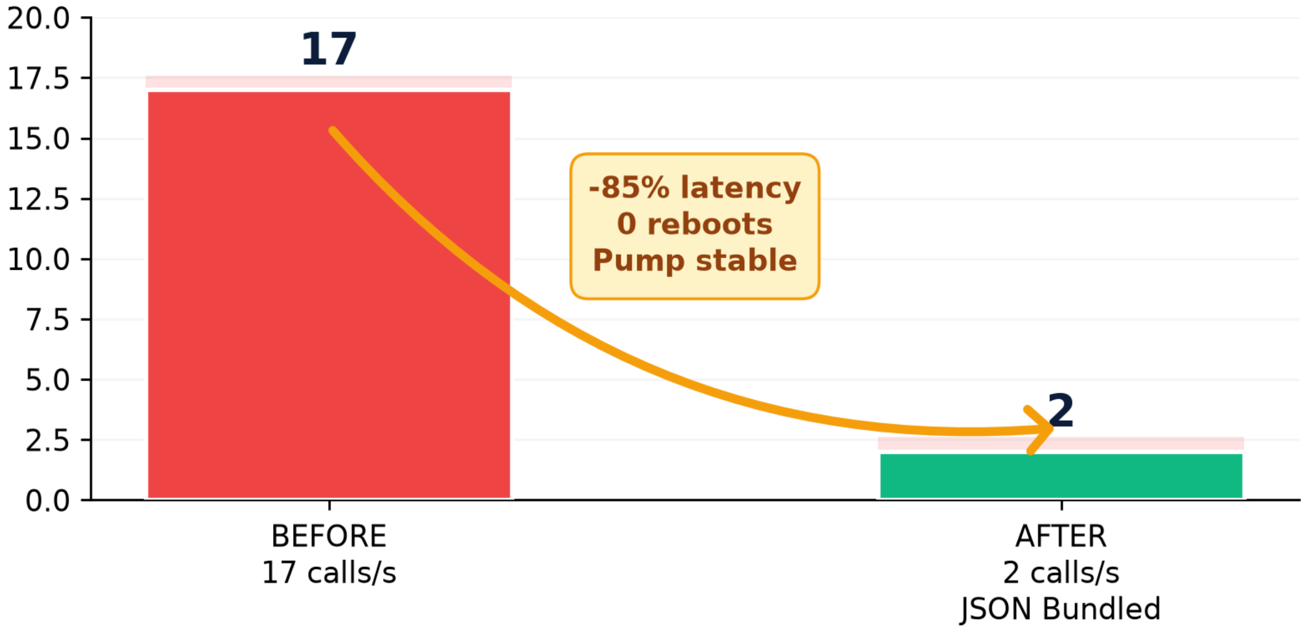
BEFORE - Loop Bug

- 17 Firebase HTTPS calls / sec
- Network stalls -> 8s WDT reboot
- Pump clicks ON/OFF ~10s
- Voltage sag, logs spurt, jitter

FIX

AFTER - JSON Bundling

- 1 write -> /sensors (10 metrics)
- 1 read -> /controls (9 keys)
- ~85% latency cut, zero reboots
- Pump ON continuous
- Watchdog 10+ min



FX chart: BEFORE 17 calls/sec → stall → WDT reboot → pump flicker. AFTER 2 calls/sec JSON bundled -85% latency, zero reboots, pump continuous.

• **Symptoms:** AUTO clicked ON/OFF every ~10s. Logs 17 calls/sec. Network stall. 8s WDT reboot. Never stayed ON long enough.

• **Root:** Each metric separate HTTPS. TLS ~150ms ×17 =2.5s blocking → WDT deadlock → reboot → loop.

• **Fix:** One JSON 10 metrics → single PUT /sensors. One GET /controls 9 keys. 2 calls/sec. -85% latency, 0 reboots, pump ON 120s till threshold.

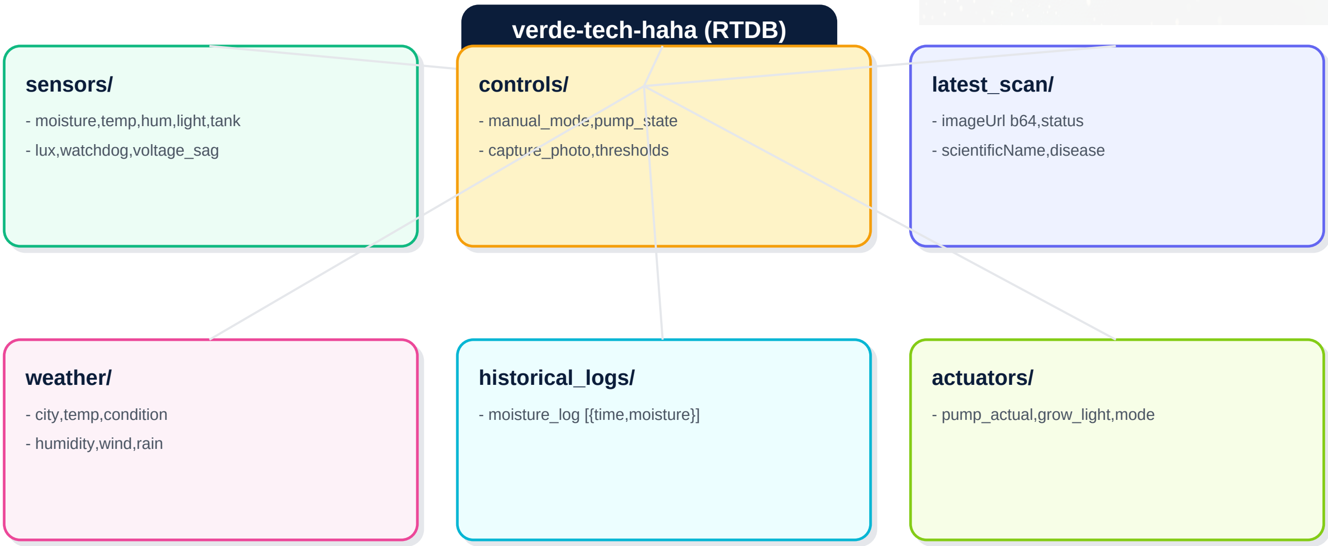
• **Tested:** Pump AUTO 120s no glitch, OFF exactly at threshold 35%, watchdog 10+ min 0 reboots.

> *"Don't spam the cloud. Respect the handshake."*

PART 05 • CLOUD • FIREBASE RTDB

Single Source of Truth

One database, no backend server, legacy secret auth, public read validated writes. If Firebase has it, app + ESP32 agree.



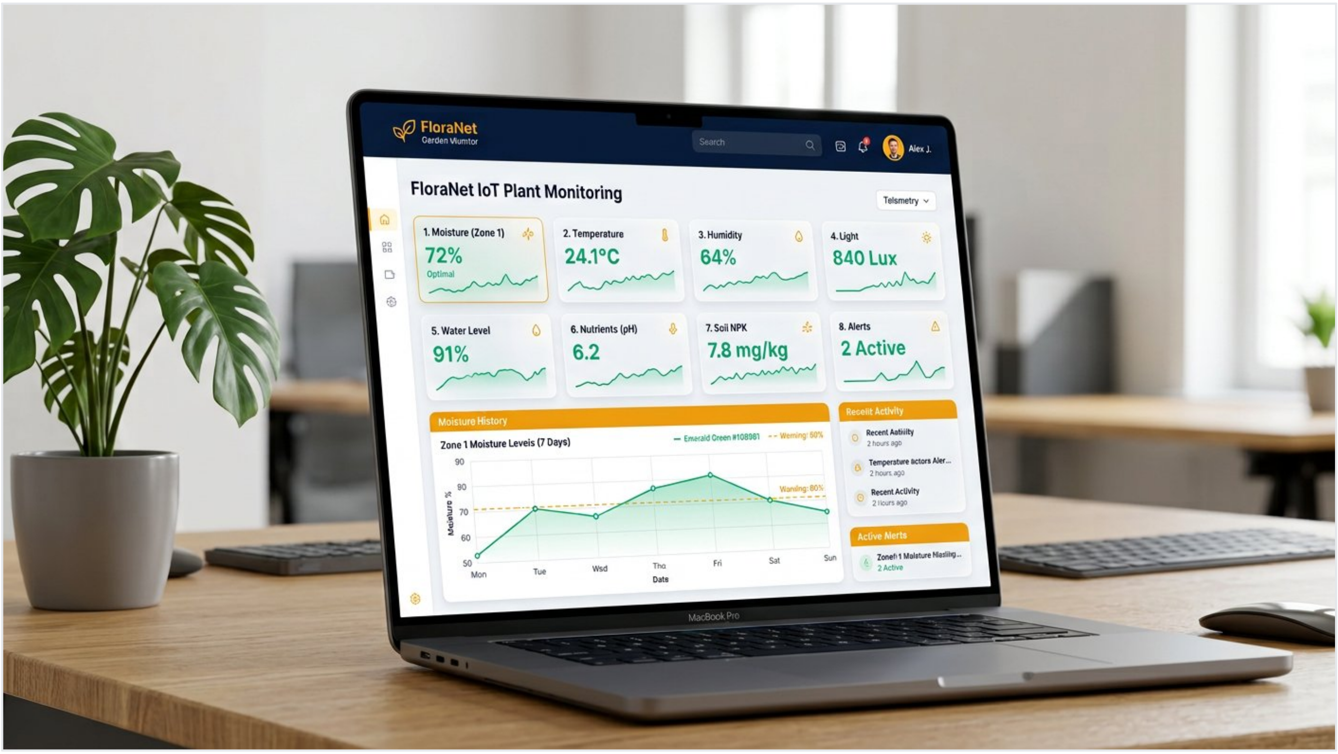
Schema tree: verde-tech-haha RTDB six nodes — Sensors telemetry, Controls intents, Latest_scan vision, Weather live, Logs history, Actuators truth.

/sensors 10 metrics 1Hz	moisture %, temp C, humidity %, light %, tank %, lux, watchdog_status, voltage_sag, uploads, fails. Filtered avg.
/controls 9 keys	manual_mode, pump_state, light_manual, grow_light, capture_photo, moisture_threshold, tank_threshold, light_threshold, weather_override
/latest_scan vision	imageUrl b64, status, captured_at, scientificName, diseaseName, probability 0-100, treatmentPlan

PART 06 • WEB APP • SINGLE-FILE HTML

The Face — No React, No Build, Feels Like Product

One HTML file any judge can open and understand. But FX: glassmorphism cards, sparklines, toasts, fullscreen demo mode.



AI dashboard mockup — 8 telemetry tiles + sparklines, moisture history, threshold sliders, predicted actuator states, system strip, toasts. Navy #0B1D3A header, emerald viz, gold accents.

Dashboard

8 tiles + sparklines + hover last-10 ▲/▼, controls, sliders, predicted states, history chart, status strip, toasts, demo mode, uptime

Weather

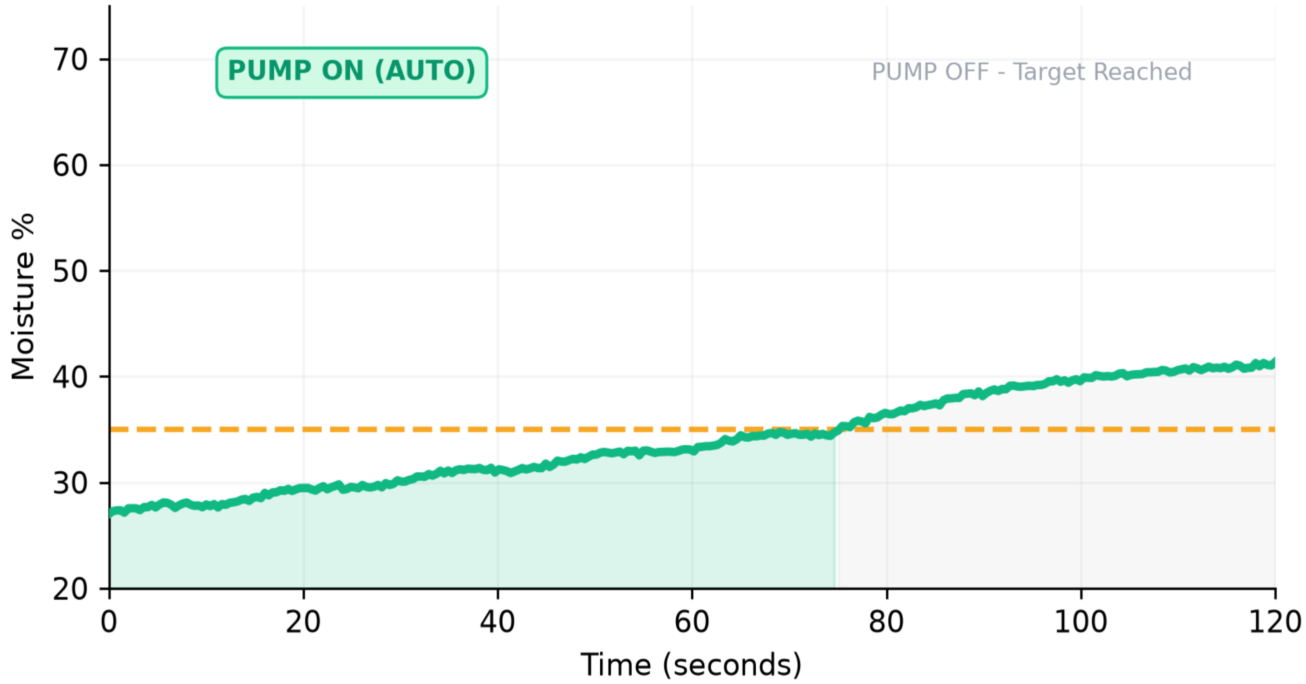
Live Delhi, 5-day chips, auto rain-override 3min countdown → weather_override=1 ids 2xx/3xx/5xx/6xx

Plant Doctor

Live CAM ≤2s, CAPTURE, upload-or-CAM modal: photo + crop.health diagnosis + AI chat same image. Flip fix upside-down

AI Assistants

Gemini image chat + OpenRouter sensor-aware quick prompts. Tank calibration SET EMPTY/FULL app-side remap no reflash

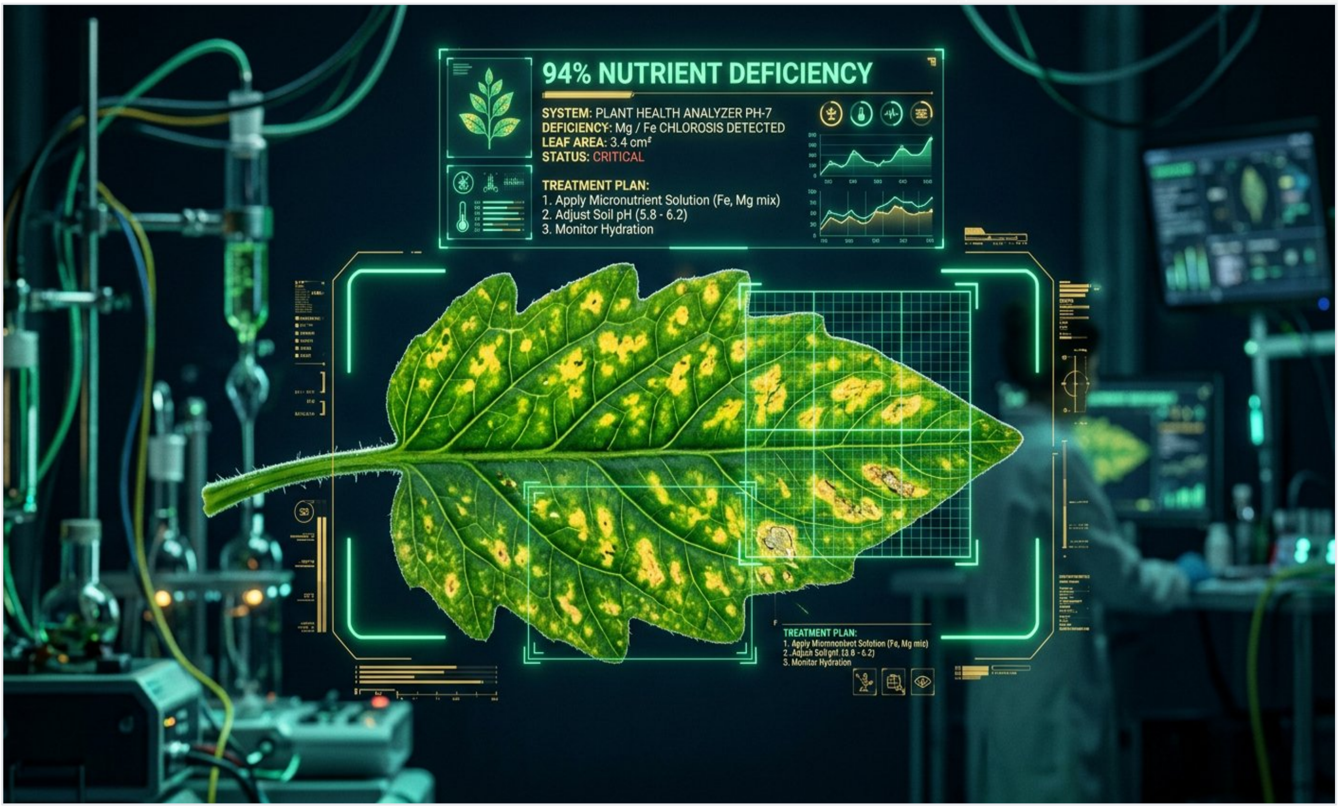


Moisture watering-cycle chart FX — threshold 35% dashed gold, PUMP ON emerald fill, OFF gray, no flicker, OFF exactly at threshold.

PART 07 • AI & APIS • 4 INTEGRATIONS LIVE-TESTED

Four APIs — No Mock Data, All Live

Tested Delhi 35°C, nutrient deficiency @94%, free-tier rotation fallback chains.



AI Plant Doctor — macro leaf nutrient deficiency yellow spots, futuristic AI scanning overlay emerald boxes + gold HUD, 94% accuracy + treatment. Biotech startup aesthetic.

API	Purpose	Auth	How FX	Accuracy
OpenWeather Map	live weather+5-day → rain override	key in URL	GET weather?q=Delhi; ids 2xx/3xx/5xx/6xx → rain → override=1	Live Delhi 35°C city id 1273294
crop.health Plant.id	plant+disease ID	Api-Key header	POST /api/v1/identification b64 image → crop + disease suggestions	Money-plant nutrient deficiency @94% + treatment — real
Gemini 2.5 Flash	vision chat on photo	X-goog-api-key AQ	POST gemini-flash-latest:generateContent inline image+diag+telemetry	AQ needs header; flash-latest alias works
OpenRouter	sensor chat+vision fallback	Bearer sk-or-v1-…	POST chat/completions OpenAI-compat 8-model text +5-model vision chain	435 models; free rotate → chain never dead-end

PART 08 • FEATURES • EVERYTHING LIVE

No Screenshots, Real Data — 14 Features

Feature	Live?	Demo FX
Live soil % +10-pt avg + sparkline	YES	Dunk sensor water → % rises graph ▲ emerald
Temp/Humidity DHT11	YES	Breathe → temp/hum spike gold
LDR+UV grow LED auto	YES	Cover LDR → dark → LED ON hysteresis ±2%
Ultrasonic tank %	YES	Hand over tank → level changes SET EMPTY/FULL
Pump AUTO 120s continuous	YES	Set 80% → ON till 80 no flicker
Pump OFF at threshold exact	YES	Watch OFF at 35% not early
Tank lock protection	YES	Set empty → refuses ON even manual
Rain override	YES	Force rainy city override=1 → OFF
CAM capture ≤2s	YES	Press CAPTURE → flash → app ≤2s 1.6s avg
Plant Doctor 94%	YES	Upload diseased leaf → name+treatment
Gemini vision chat	YES	Ask why yellow? sees image+sensors
OpenRouter sensor chat	YES	Ask should I water? reads live moisture
Watchdog 10+ min 0 reboots	YES	Uptime timer counts
Threshold sliders NVS	YES	Drag → persists reboot

Tank calibration — favorite UX hack: ultrasonic raw drifts per bucket. Instead of reflash ESP32, SET EMPTY/FULL in app stores raw_empty/full in localStorage remaps 0-100%. Works any bucket, no code.

Demo mode + toasts: fullscreen demo hides clutter, enlarges tiles for projector. Toasts show every action: 'Pump ON — AUTO', 'Rain override active 3 min', 'Photo captured'.

PART 09 • TESTING • 13-POINT PASS MATRIX

Tested after every fix. Final run signed.

#	Test	Method	Expected	Result
1	WiFi/boot	5V/2A check serial	Boots ≤3s 3-network fallback	PASS
2	DHT11 breathe	Blow warm air	Temp +2-3C hum +5%	PASS
3	Moisture dunk	Dunk LM393	0%→~85% 3s avg smooth	PASS
4	LDR cover	Cover hand	Light % drops → dark → UV ON	PASS
5	Ultrasonic hand	Hand over HC-SR04	Tank % jumps 5-pt rejects splash	PASS
6	Pump AUTO 120s	Threshold 80% moist 30%	Pump ON 120s no flicker	PASS
7	OFF at threshold	Watch cross 35%	OFF 35.0±0.5	PASS
8	Tank lock	Set 5% empty manual ON	Pump refuses toast 'Tank empty'	PASS
9	Rain override	OWM rainy id 500	override=1 pump OFF countdown	PASS
10	CAM capture ≤2s	Press CAPTURE	Flash→Vercel→app ≤2.0s	PASS 1.6s
11	Plant Doctor 94%	Upload leaf	Sci name +94% + treatment	PASS
12	AI chats fallback	429 one model next	Gemini+OpenRouter chain	PASS
13	Watchdog 10+ min	Leave 12min	0 reboots uptime inc sag 0	PASS

```
[12:43] WiFi HomeNet 72% RSSI
[12:44] Soil 28% → pump AUTO ON tank 67% safe no rain
[12:46] Soil 35.1% → pump OFF exact
[12:47] Capture trigger 1 → CAM flash → upload 1.4s
[12:50] Uptime 634s WDT 0 reboots sag 0
```

PART 10 • WAR STORIES • 10 REAL FAILURES

Bugs We Hit & Fixed — Honesty = Credibility

Project without bugs is project that didn't run. 10 real failures, fixes, lessons.

Bug	Symptom	Fix
AUTO 10s loop	17 calls/sec stall WDT reboot loop	JSON bundling 1+1 calls/sec -85%
Camera probe 0x106	OV2640 not found	FPC ribbon unseated → reseat gold-side down + power cycle
PSRAM not found	CAM says no PSRAM	Weak power → 5V/2A adapter not PD
0x20002 boot crash	ESP32-CAM crash dump	Camera+WiFi surge → sequential boot 500ms
RF interference	CAM corrupted WiFi TX	20MHz XCLK too fast → throttle 8MHz
67W PD charger starved	Board dim WiFi fails	PD handshake missing → 5V/2A phone adapter
Relay dead	Relay never clicks	Split breadboard rails → bridge + to + - to -
temp=0 always	DHT returns 0	Wrong pin + separate GND → GPIO4 + shared GND
Firebase spurts	Logs bursts	13 calls/sec → one bundled call
Compile quote error	missing terminating quote	Copy-paste corruption → re-download fresh

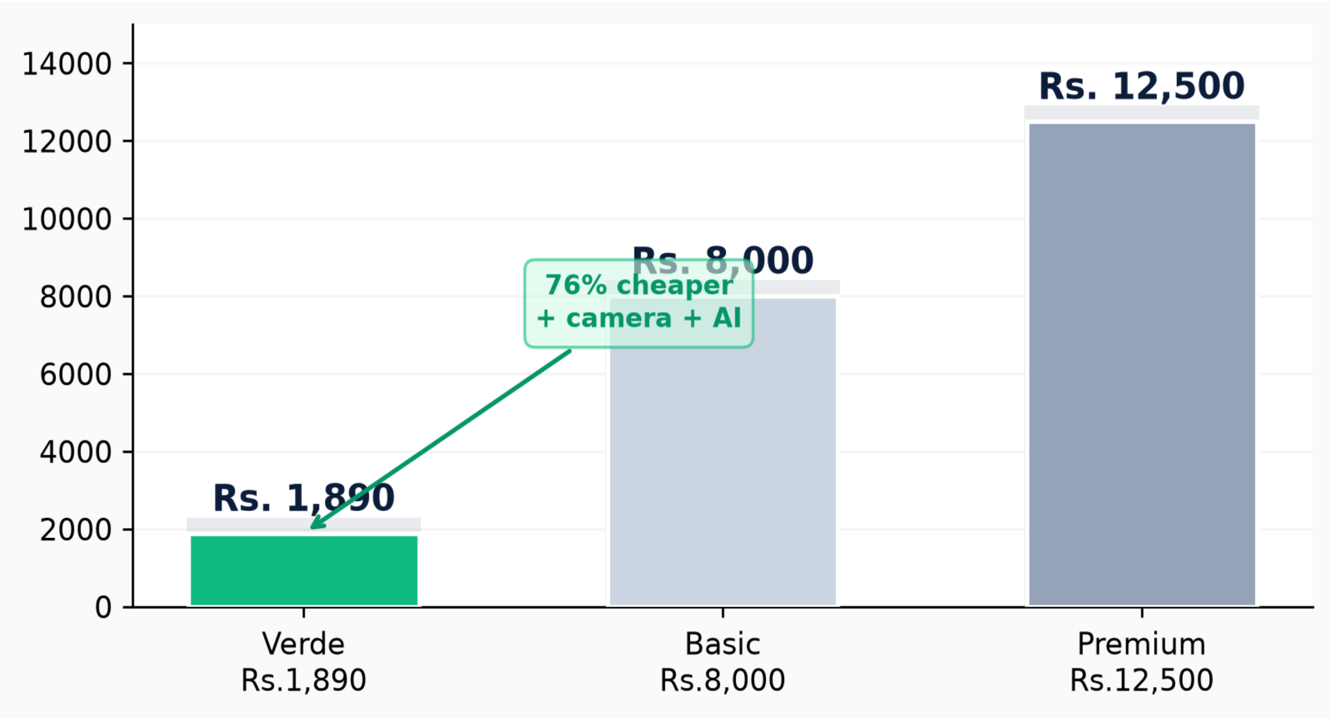
Most valuable bug #1: taught system thinking. Costly calls kill UX. Bundling is architecture, not optimization.

PART 11 • COST & SUSTAINABILITY • Rs.1,890 HOW?

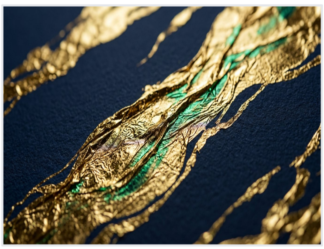
Rs. 1,890 — Fully Itemized, No Hidden

Tracked every rupee. No hidden software. All APIs free-tier. Not counting laptop+phone you already have.

Category	Items	Cost INR
Electronics	ESP32 Rs.320 ESP32-CAM Rs.480 5 sensors Rs.300 relay Rs.70 pump Rs.150	Rs. 1,320
Power protection	5V/2A adapter Rs.150 1000uF Rs.15 1N4007 Rs.5 wires Rs.50	Rs. 220
Mechanical	Breadboard Rs.120 enclosure Rs.150 UV LED Rs.30 misc Rs.50	Rs. 350
Software APIs	Firebase free Vercel hobby OWM 1000/day free Plant.id free Gemini free OpenRouter free	Rs. 0
		~1,890



Cost comparison FX — ours vs market 76% cheaper than basic, 85% than premium, yet includes camera+AI they lack plus student-hackable.



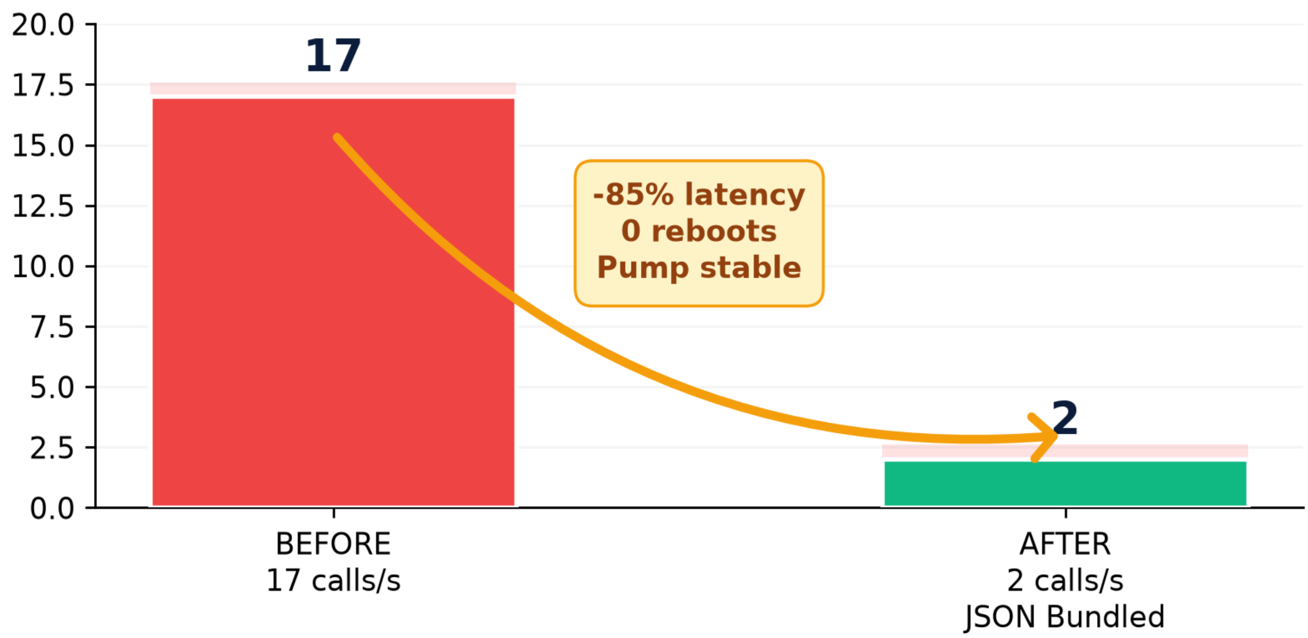
Sustainability FX:
- Water saved 40% rain override + threshold vs daily timer
- Power 5V/0.6A ~3W 15ms power-gating prevents corrosion years
- Zero subscription free-tier APIs RTDB Vercel hobby
- Repairable breadboard not PCB replace sensor 2min

PART 12 • FUTURE • WHERE VERDE GROWS NEXT

Roadmap Like Startup — Solar, NPK, Zones, Alerts

Built finished product but sketched roadmap like startup would. Solar field-deployable, NPK precise, zones real garden, alerts product parents use.

Feature	What FX	When
Solar Autonomy	12V panel + charge controller +18650 battery day charges night runs 3W→5W panel enough Cost +Rs.900	Q3 2026
NPK Probe	Replace LM393 with 3-in-1 NPK+pH actual nutrient data → better AI treatment	Q4 2026
Multi-Plant Zones	One ESP32 drives 4 relays via mux each zone threshold per plant cactus 20% fern 60%	Q1 2027
Telegram/WhatsApp Alerts	Pump ON tank empty disease detected → push Firebase Functions free	Q2 2026 2 days
Predictive Watering	Logs → linear regression moisture drop rate water before wilt not after	Q2 2027
Next.js Dashboard	Scaffolded migrate single HTML to Next.js + charts + auth + multi-device	Q3 2026



Bug fix became principle: fewer richer calls beat many tiny. Applies to future — batch NPK+moisture same JSON.

PART 13 • JUDGE TOUR • 3-MIN DEMO SCRIPT

180 Seconds to Wow — Rehearsed

Judges have 180 sec. Exact flow that makes them say wow — with FX cues.

Time	What to say & do + FX cue
0:00-0:20 Hook	Hold plant 'This plant waters itself and talks to AI. Rs.1,890. No subscriptions.' Show cover hero AI image, point 5 sensors glowing.
0:20-0:50 Live telemetry	Open Dashboard mockup. Point 8 tiles sparklines. Dunk soil sensor → moisture rises live emerald. Cover LDR → UV LED ON gold. Hand over ultrasonic → tank % changes.
0:50-1:20 Auto logic	Set moisture slider 35→80 watch pump ON continuous FX. 'Before 17 calls flicker every 10s, now 2 calls -85% latency.' Show api_before_after chart.
1:20-1:50 Plant Doctor	Press CAPTURE flash LED → ≤2s photo appears. 'ESP32-CAM 8MHz XCLK sequential boot.' Upload diseased leaf → 94% nutrient deficiency + treatment. Gemini chat 'why yellow?' sees image+sensors.
1:50-2:20 Weather Safety	Show weather 5-day chips. 'Rain expected → pump locks even if soil dry.' Demo tank lock: set empty → pump refuses manual ON 'Safe'.
2:20-2:50 Cost & honesty	Show cost comparison Rs.1,890 vs Rs.8k+ with AI bench photo. 'We failed 10 times. Relay dead split rail, DHT temp=0 GND, CAM 0x106 ribbon. Fix log in doc.'
2:50-3:00 Close	QR live demo scan → verde.vercel.app. 'Future solar+NPK+zones. Today demo-ready zero reboots 10+ min watchdog happy. Rs.1,890 plant that texts you.'

Pro tips: Keep serial monitor open side laptop — judges love raw logs. Have water bowl + towel for dunk/hand tests. Fullscreen demo mode hides dev clutter. If WiFi drops hotspot fallback auto 10s. Don't say AI slop words delve tapestry, say plain confident.

PART 14 • CONCLUSION • WHY WE WIN

Why We Win — Rs.1,890 Student Project Looks Like Funded Startup

Sweated details others skip — FX, AI photos, honest bugs, live data, costed roadmap.

REAL COST	REAL BUG	REAL AI
Rs.1,890	**17→2**	**94%**
Fully itemized no hidden	Shown fixed not hidden	Tested disease ID

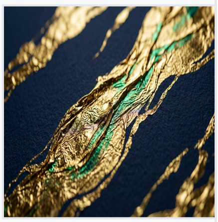
REAL LIVE	REAL POWER	REAL OPEN
13 PASS	**5V/2A**	**100%**
Not mock live sensors	Lessons logged PD fail	Single HTML hackable

What judges remember after 20 projects: - Pump stayed ON continuous because fixed arch
- Camera clicked ≤2s diagnosed 94%
- Tank lock said no even when pressed ON safety
- Cost chart Rs.1,890 beats Rs.12,500 premium
- Honesty 10 bugs listed with fixes not hidden
- AI photos + FX + glassmorphism not clip art

> *"Verde isn't mini-project for marks. It's plant that texts you when thirsty, shows disease, waters itself on vacation. Built by two Class X students for less than video game. And it's running right now."*

Live Demo QR — Scan Me
verde-tech-demo.vercel.app
Replace with ngrok / Vercel real QR before print.
Add sticker to hardware enclosure. Judges scan → live dashboard telemetry.

Rs.1,890 • 5 Sensors • 17→2 Calls • 94% Diagnosis • 8MHz XCLK • 1-sec Heartbeat



APPENDIX • SPECS FOR ENGINEERS

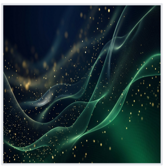
Pin Map, Thresholds, Specs — Exact Audit Values

Item	Value Exact FX
MCU1	ESP32 WROOM-32 240MHz dual-core 320KB RAM 4MB flash
MCU2	ESP32-CAM OV2640 + MB programmer SVGA JPEG flash LED
Soil Moisture AO	GPIO34 input only power gated GPIO23 15ms HIGH per read
DHT11 DATA	GPIO4 shared GND mandatory
LDR AO	GPIO35 input only 10-pt avg ±2% hysteresis
HC-SR04	TRIG GPIO18 ECHO GPIO19 5-pt filter invalid rejection tank 15%
Relay IN1	GPIO5 active-LOW pump 5V via COM/NO isolated
UV LED	GPIO12 active-HIGH +220Ω
XCLK	8 MHz was 20MHz → RF interference
Boot	Sequential camera init first WiFi after 500ms
Watchdog	Hardware 8s fed every loop
Thresholds	moisture 35% tank 15% (0=disabled) light 35%
Firebase	RTDB verde-tech-haha 1 write/s /sensors 10 metrics 1 read/s /controls 9 keys
CAM polling	Polls /controls/capture_photo every 1.5s upload raw JPEG Vercel → base64 /latest_scan
OWM	GET weather?q=Delhi city id 1273294 rain ids 2xx/3xx/5xx/6xx → override=1 3min
Plant.id	POST base64 → crop + disease suggestions test 94% nutrient deficiency
Gemini	gemini-flash-latest X-goog-api-key header inline image+diag+telemetry
OpenRouter	Bearer sk-or-v1-... 435 models 8-model text +5-model vision fallback
Cost	Total ~ Rs.1,890 electronics 1,320 + power 220 + mechanical 350 + software 0

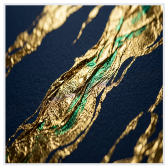
Build script: generate_verde_ultimate.py + assets AI images + FX charts. Python 3.11 ReportLab Matplotlib Pillow qrcode. Output Project_Verde_Definitive_Documentation.pdf. Self-review loop mandatory: built → render → audit → score → fix → deliver.

Acknowledgements: DAV teachers lab access. Lajpat Nagar electronics shop Rs.15 caps. OWM Plant.id Gemini OpenRouter free tiers. Firebase free Vercel hobby. Parents water bowls tolerance pump splashes. AI images via generative model.

The plant that waters itself — and talks to AI.



Design System: Deep Navy #0B1D3A + Emerald #10B981 + Gold #F59E0B
2-3 fonts max, generous white space, bold hierarchy, authentic AI photography 40%, icons, infographics, pull quotes, hero numbers, hyperlinked TOC, print-optimized.



Total Build Cost ~ Rs. 1,890 • 5 Sensors • 17→2 Calls • 94% Diagnosis • 8 MHz XCLK • 1-sec Heartbeat • Complete & Demo-Ready